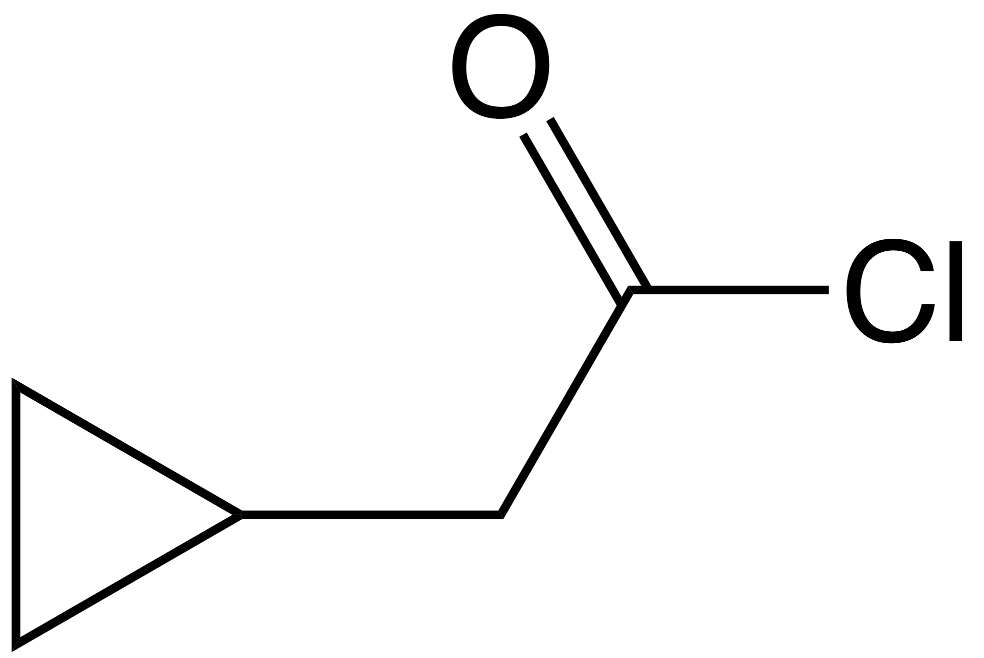
# Boiling Point Challenge

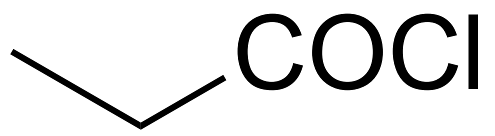



**What is the boiling point?**

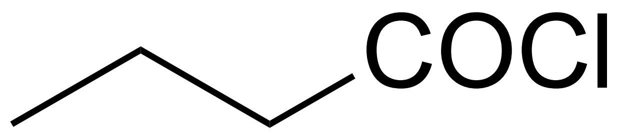
A: -10 – 0 °C
B: 0 – 20 °C
C: 20 – 40 °C
D: 40 – 60 °C
E: 60 – 80 °C
F: 80 – 100 °C
G: 100 – 120 °C
H: 120 – 140 °C
I: 140 – 160 °C
J: 160 – 180 °C
K: 180 – 200 °C
L: > 200 °C

# Boiling Point Challenge (with Hints)

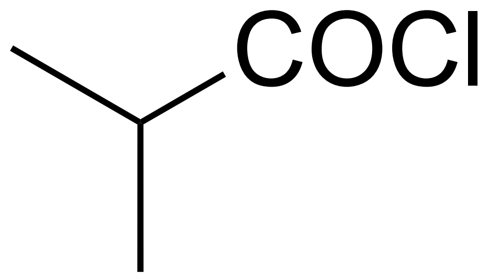
52 ºC



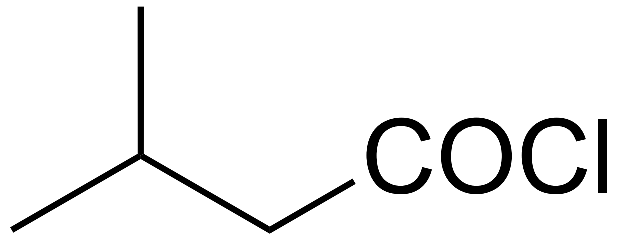
80 ºC



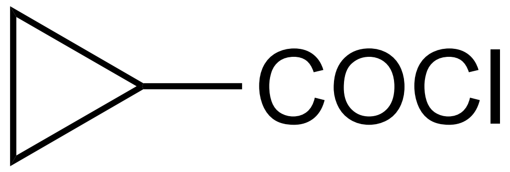
102 ºC



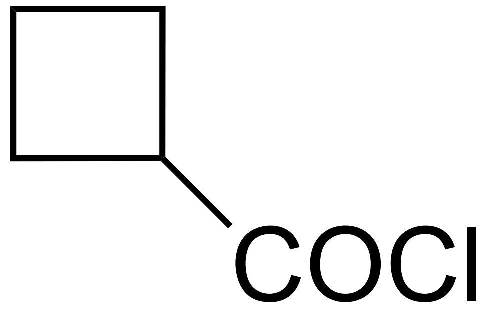
93 ºC



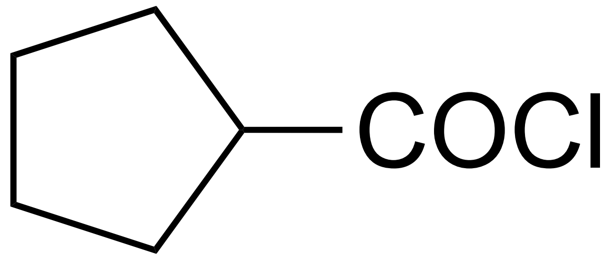
116 ºC



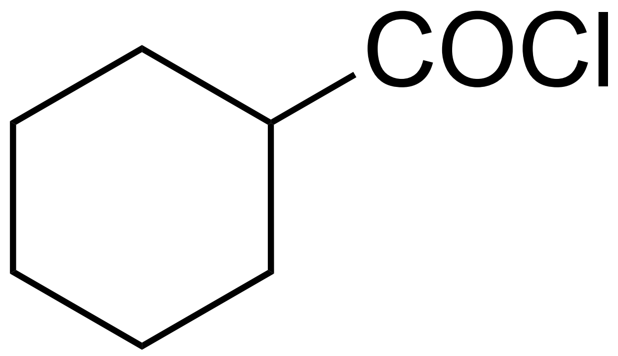
105 ºC



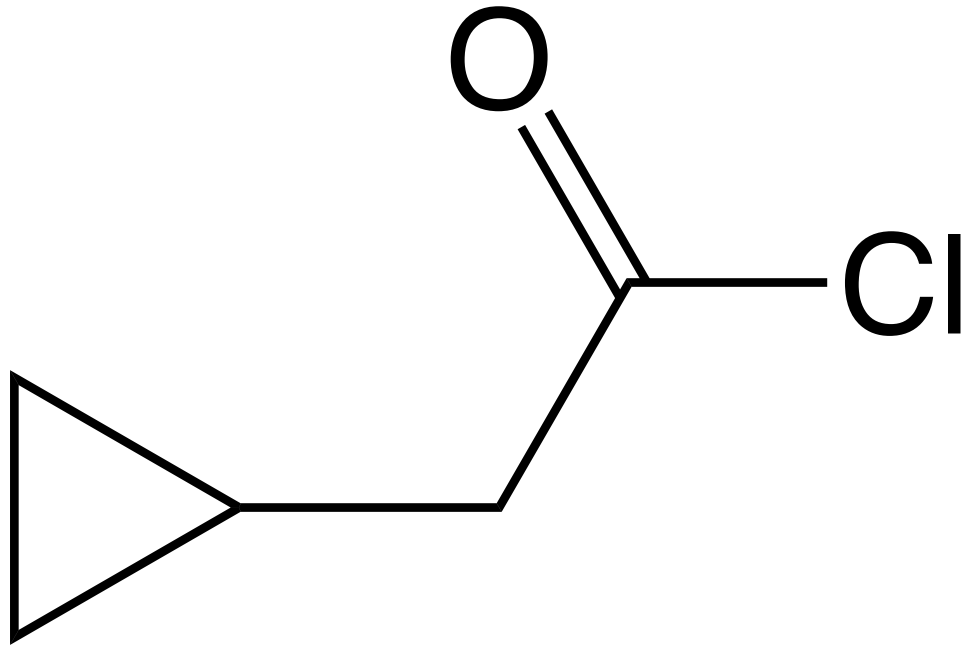
143 ºC



161 ºC



185 ºC



**What is the boiling point?**

**F: 80 – 100 ºC**

**G: 100 – 120 ºC**

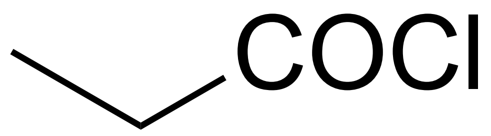
**H: 120 – 140 ºC**

**I: 140 – 160 ºC**

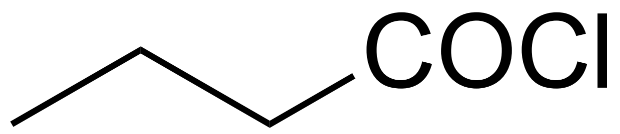
**J: 160 – 180 ºC**

# Answer

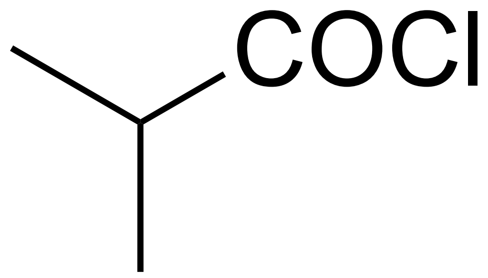

52 ºC



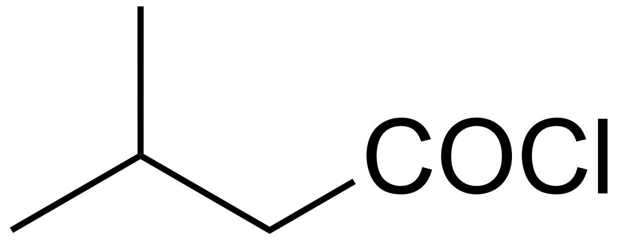

80 ºC



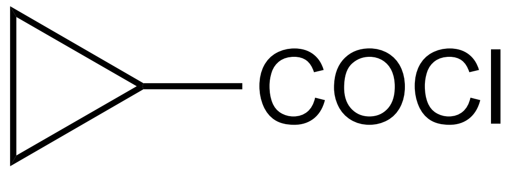

102 ºC



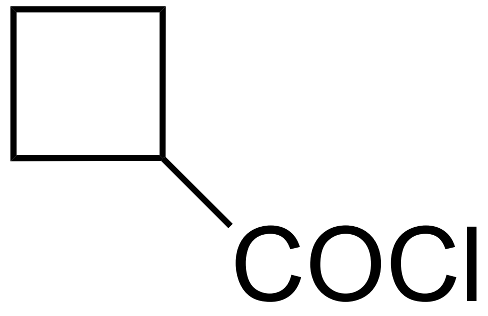

93 ºC



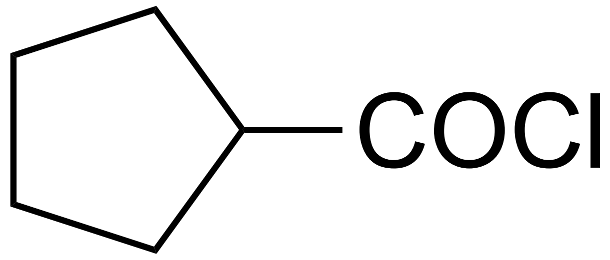

116 ºC



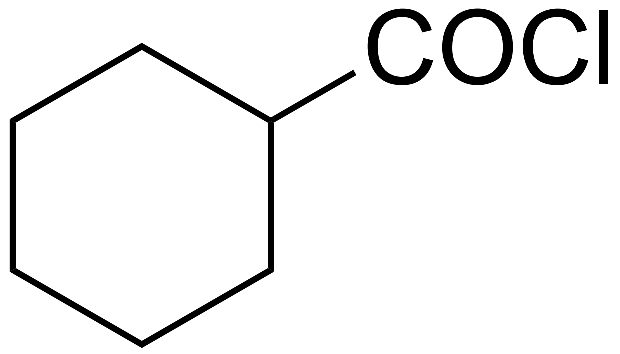

105 ºC



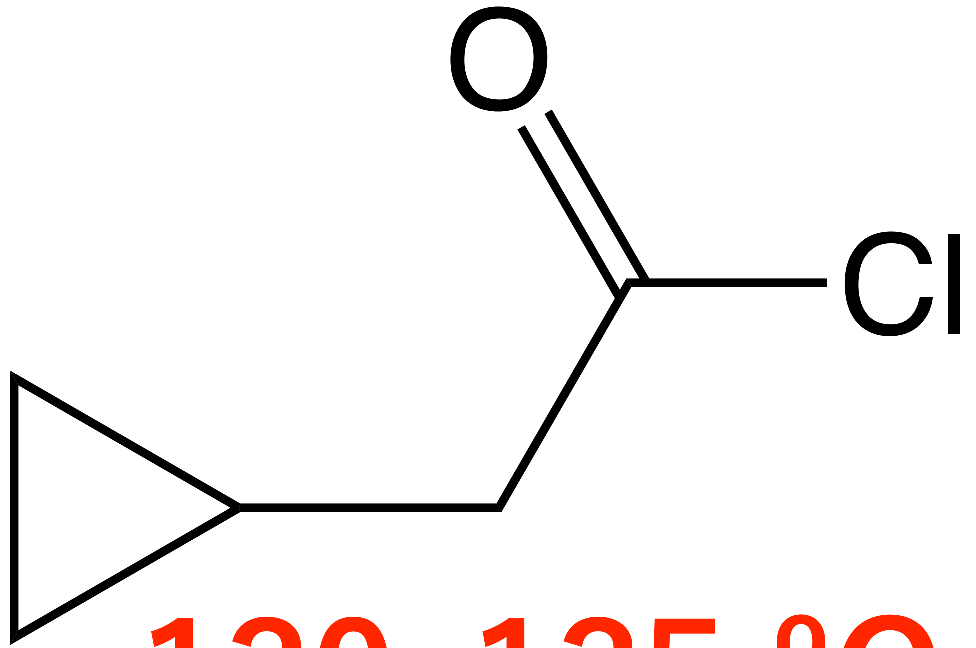

143 ºC




161 ºC




185 ºC




**130–135 ºC**

**F: 80 – 100 ºC**

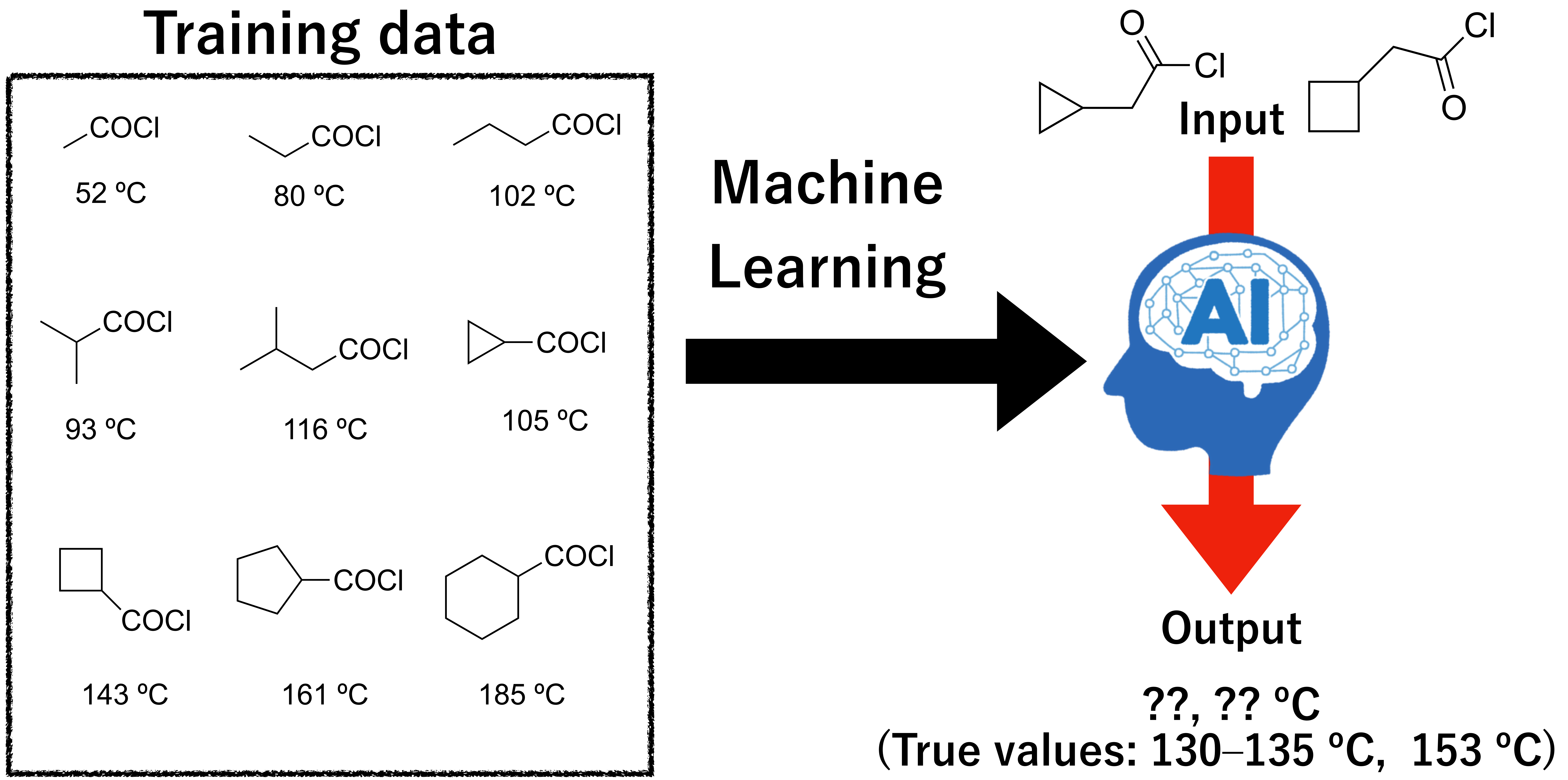
**G: 100 – 120 ºC**

**H: 120 – 140 ºC**

**I: 140 – 160 ºC**

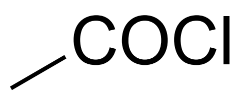
**J: 160 – 180 ºC**

# Boiling Point Prediction Using Machine Learning

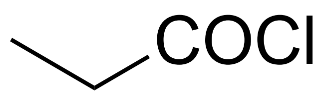



**Can you design molecular descriptors that help the model predict boiling points accurately?**

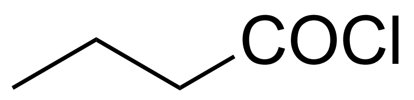
# Creating Molecular Descriptors for Boiling Point Prediction



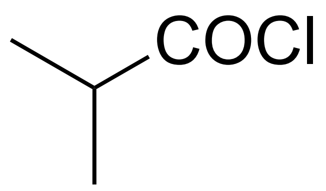
52 ºC



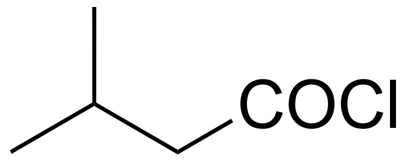
80 ºC



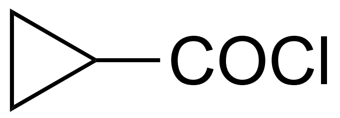
102 ºC



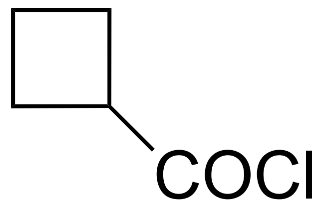
93 ºC



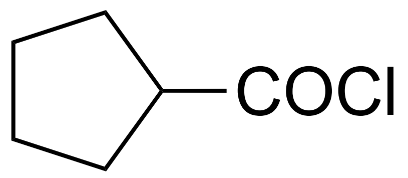
116 ºC



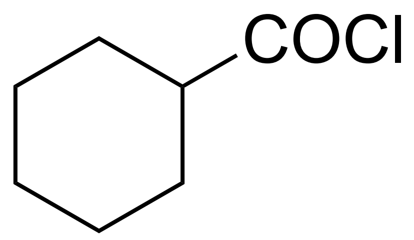
105 ºC



143 ºC



161 ºC



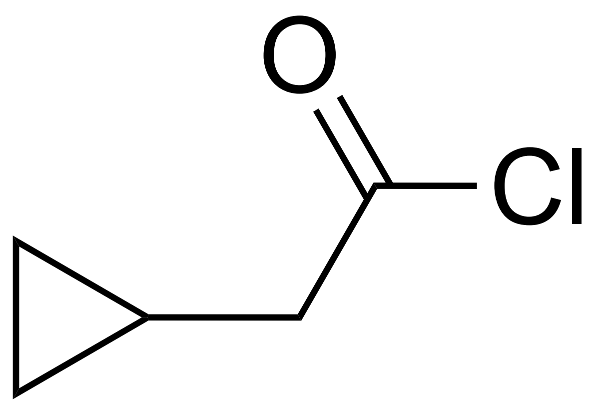
185 ºC

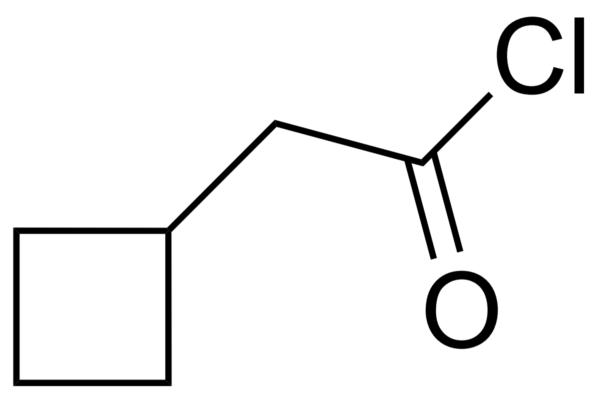
## Molecular weight and counts of structural features (CH3, CH2, CH, and rings)

| MW | CH3 | CH2 | CH | Cyc3 | Cyc4 | Cyc5 | Cyc6 | BP |
|---|---|---|---|---|---|---|---|---|
| 78.5 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 52 |
| 92.52 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 80 |
| 106.55 | 1 | 2 | 0 | 0 | 0 | 0 | 0 | 102 |
| 106.55 | 2 | 0 | 1 | 0 | 0 | 0 | 0 | 93 |
| 120.58 | 2 | 1 | 1 | 0 | 0 | 0 | 0 | 116 |
| 104.53 | 0 | 2 | 1 | 1 | 0 | 0 | 0 | 105 |
| 118.56 | 0 | 3 | 1 | 0 | 1 | 0 | 0 | 143 |
| 132.59 | 0 | 4 | 1 | 0 | 0 | 1 | 0 | 161 |
| 146.61 | 0 | 5 | 1 | 0 | 0 | 0 | 1 | 185 |

**Complete the table in Excel and save it as “BP.csv” (CSV format).**

# Preparing Input Data for Prediction




**BP_pred.csv**

| MW | CH3 | CH2 | CH | Cyc3 | Cyc4 | Cyc5 | Cyc6 |
|---|---|---|---|---|---|---|---|
| 118.56 | 0 | 3 | 1 | 1 | 0 | 0 | 0 |
| 132.59 | 0 | 4 | 1 | 0 | 1 | 0 | 0 |

**Complete the table in Excel and save it as “BP_pred.csv”.**

## Let's Build a Machine Learning Model in Google Colab!